Lynx and Hare Model Formulation and Scaling

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1 Problem and observation convention

The project asks whether a mechanistic predator–prey model can explain recurring hare–lynx fluctuations and support useful predictions beyond the observations used for calibration. This section establishes the equations, their biological assumptions, units and dimensionless form. It supplies the common mathematical foundation for the team's stability analysis, numerical simulations and parameter estimation.

The completed historical iteration uses annual observations labelled 1900–1920. The Stan case study supplies a public transcription of Hudson's Bay Company pelt records in thousands of pelts, with a documented source chain [2]. These measurements are trade records, not population censuses. We interpret $H(t)$ and $P(t)$ as nonnegative hare and lynx abundance *indices*, respectively, in explicitly declared species-specific units. The comparison between a continuous state and an annual pelt total is an observation approximation: trapping effort, capture probability, seasonal aggregation and unmodelled ecology can all affect the relationship.

Advay's frozen data split and the reference configuration use one index unit per thousand pelts for each species [1, 8]. The Stan compilation remains the source for this iteration. The alternative tsibbledata compilation disagrees with it, a documented source limitation; neither series establishes absolute animal abundance, and they are not combined. Set $t = \text{Year} - 1900$ in years. The implemented evaluation design uses 1900–1915 for calibration and 1916–1920 for final scoring. Normalization and parameter selection use training observations only. Forecasts have already been frozen and scored; this section documents their observation convention without changing it. It does not provide additional historical fits.

1.1 Explicit observation model

For year Year_i and $t_i = \text{Year}_i - 1900$, write

$$Y_{H,i} = H(t_i) + \varepsilon_{H,i}, \qquad Y_{P,i} = P(t_i) + \varepsilon_{P,i}. \tag{1}$$

Here Y is the recorded annual pelt count in thousands; H, P are the model's noiseless pelt indices; and ε is observation discrepancy in the same species-specific units. Under the working proportional-index assumption O1,

$$H(t) = c_H N_H(t), \qquad P(t) = c_P N_P(t), \qquad c_H, c_P > 0, \tag{2}$$

where N_H, N_P denote latent animal abundance and the unknown observation factors are constant. We fit the effective equations directly in pelt units: *no observation-scale factor is fitted.* This does not set $c_H = c_P = 1$ or identify actual animal counts. Constant factors are absorbed into the effective coefficients through the transformation derived in Eq. (26).

Each annual total is approximated by a point observation of the state, as in the implemented convention [1, 8]. Thus H, P have units of thousands of pelts, and $\dot{H}, \dot{P}$ have units of thousands of pelts per year. No seasonal observation integral is fitted. A latent-population model with estimated observation factors, or an integral over the trapping season, would represent additional mechanisms but require further assumptions and unknowns. O1 and the point-sample approximation keep the present comparison tractable, at the cost of limited biological interpretation.

The main observation loss compares additive residuals after division by the training maxima $s_H = 77.4$ and $s_P = 59.4$:

$$R_{H,i} = \frac{H(t_i) - Y_{H,i}}{s_H}, \qquad R_{P,i} = \frac{P(t_i) - Y_{P,i}}{s_P}. \tag{3}$$

For the PINN data term the state in this expression is the network approximation; for least squares it is the integrated ODE state. Both methods use the same species scales. These weights are a declared comparison convention, not measured noise standard deviations. Additive discrepancy notation alone establishes neither Gaussian noise nor independence of observations. The reported log-error sensitivity instead emphasizes relative discrepancies. A separate sensitivity fits the 1900 initial states within $[0.5, 1.5]$ times their observed values [8]; the two checks change different assumptions and remain separate outcomes.

2 From Lotka–Volterra to the chosen model

2.1 The reference model and its limitation

For a first comparison, let

$$\dot{H} = rH - cHP, \qquad \dot{P} = ecHP - mP, \tag{4}$$

where a dot means differentiation with respect to years. The prey grows at rate rH without predators, encounters remove prey at rate cHP, conversion of consumed prey contributes $ecHP$ to predator growth, and predator mortality removes mP. Here $c > 0$ is the encounter coefficient; it is not the saturation parameter a introduced below. The other parameters are positive.

The positive equilibrium is $H_* = m/(ec)$, $P_* = r/c$. To see why the cycles are neutral, rather than simply asserting it, define for positive H, P

$$V = ec\,[H - H_* - H_* \log(H/H_*)] + c\,[P - P_* - P_* \log(P/P_*)]\,. \tag{5}$$

The logarithms have dimensionless arguments. Differentiation gives

$$\dot{V} = (ec - m/H)\dot{H} + (c - r/P)\dot{P} \tag{6}$$

$$= (ecH - m)(r - cP) + (cP - r)(ecH - m) = 0. \tag{7}$$

Each bracket in V has a strict minimum at its equilibrium value and diverges at zero or infinity. Thus nonzero level sets are closed curves in the positive quadrant, surrounding the equilibrium. The vector field has no other positive equilibrium on a level set, so trajectories follow periodic orbits. A perturbation generally moves the state to a different orbit; it does not relax back to one selected cycle. The equilibrium is a centre, not an attractor. Also, predator-free prey grows exponentially without a resource limit. These are reasons to refine the model; they do not mean that Lotka–Volterra cannot oscillate.

2.2 Resource limitation and saturating predation

Let the prey's per-capita net growth decrease linearly from r at low abundance to zero at $H = K$. Multiplication by H gives logistic growth,

$$G(H) = rH(1 - H/K). \tag{8}$$

The carrying capacity K describes the prey-only resource limit under fixed environmental conditions; it is not an absolute ceiling on every trajectory.

Let $g(H)$ be consumption per predator. A simple search-and-handling argument assumes an encounter-limited rate qH while searching and a maximum processing rate a. If the fractions of time searching and handling are $g/(qH)$ and g/a, their sum is one. For $H > 0$,

$$\frac{g}{qH} + \frac{g}{a} = 1 \quad \Longrightarrow \quad g(H) = \frac{aqH}{a + qH} = \frac{aH}{b + H}, \qquad b = \frac{a}{q}. \tag{9}$$

Extend continuously by $g(0) = 0$. This is the Holling type II response [3]. It assumes an effectively constant prey level over the short feeding interval and no predator interference. In index units it is an effective response law, not a direct measurement of handling time. The parameters have transparent interpretations:

$$g(b) = a/2, \qquad \lim_{H\to\infty} g(H) = a, \qquad g(H) \simeq (a/b)H \quad \text{when } H/b \ll 1. \tag{10}$$

The auxiliary $q = a/b$ has units inverse lynx index per year. It is not an additional parameter to estimate. The linear response would be simpler but cannot represent saturation; a type III response would introduce a different low-prey mechanism that the present project does not establish.

2.3 Dimensional Rosenzweig–MacArthur equations

Combining logistic growth, total prey consumption $g(H)P$, conversion and mortality gives the project's Rosenzweig–MacArthur model [7]:

$$\frac{dH}{dt} = rH\left(1 - \frac{H}{K}\right) - \frac{aHP}{b + H}, \tag{11}$$

$$\frac{dP}{dt} = \frac{eaHP}{b + H} - mP. \tag{12}$$

These are exactly the dimensional equations in the shared repository [5]. Logistic growth and saturation allow stability to depend on parameters. Whether a specified case approaches an equilibrium or exhibits a stable cycle requires the subsequent analysis and numerical evidence; it does not follow merely from naming the model.

3 Units and assumptions

Let U_H and U_P denote the chosen hare and lynx index units. Keeping these labels distinct prevents an unjustified interpretation of conversion efficiency.

Symbol	Meaning	Units
t	Elapsed time	year
H, P	Hare and lynx states	U_H, U_P
r	Low-density hare net growth rate	year^{-1}
K	Effective prey-only carrying-capacity index	U_H
a	Effective saturated consumption coefficient	$U_H U_P^{-1}$ year^{-1}
b	Prey index at half saturation	U_H
e	Effective predator conversion coefficient	$U_P U_H^{-1}$
m	Predator mortality rate	year^{-1}
H_0, P_0	States at the initial time	U_H, U_P

All six parameters are strictly positive. For example, $[aHP/(b+H)] = U_H/\text{year}$ and $[eaHP/(b+H)] = U_P/\text{year}$, matching the two left-hand sides. In separately scaled abundance indices, e is an effective conversion factor. It need not be a dimensionless biomass efficiency bounded above by one. For this historical iteration U_H means thousands of hare pelts and U_P thousands of lynx pelts. The rates r, m retain their growth/mortality interpretation only conditional on O1 and an adequate dynamical model; K, b, a, e are effective pelt-unit coefficients. Scale invariance does not establish accurate recovery of any biological parameter.

The simplifying assumptions and their consequences are:

1. A closed two-species system has no immigration, emigration, alternative prey or other predators. This isolates the proposed feedback but omits a food web.

2. Continuous, spatially mixed states omit age structure, territories and local differences. The approximation is unsuitable for individual-level extinction claims.

3. Constant parameters omit seasons, weather and changing resource availability. Fitted rates summarize the chosen interval rather than universal ecological constants.

4. Resource limitation is represented by logistic prey growth. Plants are not modelled as a third dynamical state.

5. Predation follows the type II response with no interference between predators. Predator growth depends on prey; without prey, P decays at rate m.

6. Pelt totals are treated as noisy abundance proxies. Constant observation scales are a working assumption. Unknown or changing capture rates can confound parameter inference. Harvest is not modelled as an additional removal process.

3.1 Basic consistency checks

The equations have the form $\dot{H} = HA(H,P)$, $\dot{P} = PB(H)$ with continuous coefficients for $H \geq 0$. Along a finite solution interval,

$$H(t) = H_0 \exp\left(\int_0^t A(H(s),P(s))\,ds\right), \quad P(t) = P_0 \exp\left(\int_0^t B(H(s))\,ds\right). \tag{13}$$

Positive initial states remain positive, and either zero boundary is invariant by uniqueness. A numerical method can still introduce negative approximations, so its outputs must be checked. Also $\dot{H} \leq rH(1-H/K)$, giving $H(t) \leq \max\{H_0, K\}$ for $t \geq 0$. Since $\dot{P} \leq eaP$, predator abundance cannot blow up in finite time. These checks establish basic consistency, not ecological accuracy or long-term stability.

4 Non dimensionalisation

4.1 Choosing the scales

For a fixed parameter vector define

$$s = rt, \qquad x = H/K, \qquad y = \frac{aP}{rK}. \tag{14}$$

The corresponding reference time and states are $1/r$, K and rK/a. Their units are year, U_H and U_P. The scales therefore make all three new variables dimensionless. Choosing K for prey makes the prey-only capacity one; choosing rK/a for predators makes the prey-loss coefficient one. This is the same notation used in Joel's existing dimensionless analysis [4].

Define the three dimensionless groups

$$\beta = b/K, \qquad \alpha = ea/r, \qquad \delta = m/r. \tag{15}$$

They compare half saturation with carrying capacity, maximum predator growth with prey growth, and predator mortality with prey growth, respectively. This scaling also means a sweep of K changes both β and the state scales. If the dimensional initial states are held fixed in that sweep, their scaled values must be recomputed for each K.

4.2 Substitution and chain rule

Because $H = Kx$, $P = (rK/a)y$ and $ds/dt = r$,

$$\frac{dH}{dt} = rK\frac{dx}{ds}, \qquad \frac{dP}{dt} = \frac{r^2K}{a}\frac{dy}{ds}. \tag{16}$$

Substitution into the prey equation gives

$$rK\frac{dx}{ds} = rKx(1-x) - \frac{rKxy}{b/K + x}. \tag{17}$$

Substitution into the predator equation gives

$$\frac{r^2K}{a}\frac{dy}{ds} = \frac{erKxy}{b/K + x} - \frac{mrK}{a}y. \tag{18}$$

Dividing by the respective factors yields

$$\frac{dx}{ds} = x(1-x) - \frac{xy}{\beta + x}, \tag{19}$$

$$\frac{dy}{ds} = \alpha\frac{xy}{\beta + x} - \delta y. \tag{20}$$

The two systems describe the same trajectories after the changes of units. Reducing six dimensional parameters to three groups does not prove that all six can be identified from data: physical scales and initial conditions also carry information.

4.3 Initial conditions and conversion back

At year 1900, $s = 0$, $x_0 = H_0/K$ and $y_0 = aP_0/(rK)$. Recover the dimensional variables using

$$\text{Year} = 1900 + s/r, \qquad H = Kx, \qquad P = (rK/a)y. \tag{21}$$

If r, K, a are separately specified, recover $b = \beta K$, $e = \alpha r/a$ and $m = \delta r$. The three dimensionless groups alone cannot reconstruct all six dimensional parameters.

For an arithmetic check, use the repository's synthetic values [5]:

$$(r, K, a, b, e, m) = (1, 60, 2, 20, 0.6, 0.5), \qquad (H_0, P_0) = (35, 9).$$

These give

$$(\beta, \alpha, \delta) = (1/3, 1.2, 0.5), \qquad (x_0, y_0) = (7/12, 0.3). \tag{22}$$

At this initial state, $(\dot{H}, \dot{P}) = (413/132, 261/110)$ in their respective index units per year. The dimensionless derivatives are $(dx/ds, dy/ds) = (413/7920, 87/1100)$; multiplying by $rK = 60$ and $r^2K/a = 30$ recovers the dimensional values. These are synthetic checks, not fitted historical estimates.

An equally valid alternative is $X = H/b = x/\beta$ and $Y = aP/(rb) = y/\beta$, with the same time s. Writing $\kappa = K/b = 1/\beta$ gives $X_s = X(1 - X/\kappa) - XY/(1 + X)$ and $Y_s = \alpha XY/(1 + X) - \delta Y$. This keeps state scales fixed during a K sweep when other parameters remain fixed. We use the K-based form as the main convention to match the existing team analysis, while retaining the alternative as an explanation of the scaling tradeoff.

5 Compatibility with data scaling and the PINN

5.1 Keep mathematical and numerical coordinates explicit

The historical PINN [8] uses fixed training maxima s_H, s_P, normalized outputs $u = H/s_H$, $v = P/s_P$, training duration $T = 15$ years, and internal time input $\xi = 2t/T - 1$. It differentiates with respect to elapsed years t. These numerical choices can coexist with the dimensionless theory coordinates. In elapsed time,

$$\frac{du}{dt} = ru(1 - s_H u/K) - \frac{as_P uv}{b + s_H u}, \tag{23}$$

$$\frac{dv}{dt} = \frac{eas_H uv}{b + s_H u} - mv. \tag{24}$$

The bridge to the theoretical variables is

$$x = (s_H/K)u, \qquad y = \frac{as_P}{rK}v, \qquad s = \frac{rT}{2}(\xi + 1). \tag{25}$$

If differentiation is instead performed with respect to ξ, the chain rule requires $du/dt = (2/T)du/d\xi$, and likewise for v. Thus a residual formed in years is $R_u = (2/T)u_\xi - f_H/s_H$. If one forms a residual in ξ coordinates, $\widetilde{R}_u = u_\xi - (T/2)f_H/s_H = (T/2)R_u$. With the same averaging convention, its squared penalty is $(T/2)^2$ larger; the loss coefficient must be multiplied by $(2/T)^2$ to preserve the same objective. This is a unit change, not evidence that one trained model is better.

Because all six coefficients, including r and a, are inferred in this iteration, directly using rt and $aP/(rK)$ as fixed training inputs or targets would introduce parameter-dependent coordinates. Retaining his fixed training normalization avoids that complication. The dimensionless form is for derivation, interpretation and consistent mapping, and does not require rewriting the PINN.

5.2 Why measurement units change parameter meanings

For any constant positive changes $\widetilde{H} = c_H H$, $\widetilde{P} = c_P P$, keeping time in years preserves the model form if

$$\widetilde{r} = r, \quad \widetilde{m} = m, \quad \widetilde{K} = c_H K, \quad \widetilde{b} = c_H b, \quad \widetilde{a} = \frac{c_H}{c_P}a, \quad \widetilde{e} = \frac{c_P}{c_H}e. \tag{26}$$

This follows by substituting into both derivatives, including the denominator. It leaves $\beta, \alpha, \delta, x, y$ unchanged. For example, converting both species from counts to thousands uses $c_H = c_P = 10^{-3}$: K and b divide by 1000, while a and e keep their numerical values. Unequal species scales change a and e too. Unknown pelt-to-population multipliers have the same algebraic effect, but their unknown values prevent conversion to absolute biological rates. Time-varying multipliers would add derivative terms and are outside this model.

The same transformation leaves Joel's stability ratio $\sigma = K/K_H$ unchanged, where $K_H = b(ea + m)/(ea - m)$ and positive coexistence is applicable [4]. Thus the dimensionless groups have a model-based interpretation independent of constant observation scaling. This does not confirm that the real ecosystem follows the model or has a stable limit cycle.

With the observed predator scale fixed, an arbitrary rescaling of P cannot preserve the same observations. Moreover, a occurs separately in the prey equation while ea occurs in the predator equation. Therefore the scale ambiguity alone does not prove that effective a and e are inseparable. It also does not prove that all six coefficients are well determined by 16 annual pairs.

5.3 Reference settings for the completed iteration

The reference is `configs/historical.json`, rather than the earlier synthetic configuration [8]. State order is (H, P) and parameter names are (r, K, a, b, e, m), matching the shared right-hand side. All six coefficients are fitted. Their independent positive bounds are

$$r, m \in [0.05, 3], \quad K \in [20, 500], \quad a \in [0.05, 10], \quad b \in [0.5, 300], \quad e \in [0.02, 5].$$

The units are those in the table above. These are exploratory numerical bounds, not measured ecological ranges. The completed iteration imposes neither $e \leq 1$ nor a coexistence constraint. This describes the recorded computation, rather than a new assertion of identifiability or a proposed change to the inverse problem.

Historical fitting and ODE forecasts use `model.solve`: RK45 with relative tolerance 10^{-8}, absolute tolerance 10^{-10} and maximum step 0.1 years. Rishi's tighter settings for illustrative long-run simulations are a separate calculation, not the settings of this historical comparison [8, 6].

In the main analysis the ODE starts at the observed 1900 index state $(30, 4)$. The PINN enforces its initial state with a soft penalty and reports the discrepancy; we do not describe this as an exact network constraint. In the fitted-initial-state sensitivity, each method forecasts from its fitted state at 1900. Every forecast integrates from 1900 through 1920 with frozen parameters; none restarts from a test-year observation. The historical handoff records both sensitivity variants and preserves them alongside the main result [8].

6 Validation responsibilities and limits

The derivations supply a model to test, not a claim that a published model is novel or that a PINN earns bonus marks automatically. The independent contribution here is the explicit reasoning, unit audit, coordinate transformations and interpretation under the chosen observation convention. The accompanying notebook checks algebraic equivalence at multiple exact rational inputs. These checks do not replace empirical validation or establish parameter identifiability.

Joel's contribution supplies equilibrium, stability and Hopf analysis [4]; Rishi's supplies numerical checks and parameter-regime illustrations [6]. Advay's data records and Yash's historical fits, matching least-squares benchmark and evaluation are available [1, 8]. Their empirical results belong in the corresponding report sections. Comparisons use the same training years, parameter bounds, initial-state convention and solver settings, with separate hare and lynx MAE and RMSE

in thousands of pelts. The PINN's soft initial penalty and its network–ODE discrepancy must remain visible qualifications.

Five held-out years offer limited evidence. Synthetic recovery generated by the same equations checks the inverse procedure under favourable assumptions; it is not independent evidence that those equations describe real ecology. A small training error does not show unique biological parameters. Any substantive change after the completed evaluation must be documented as a later revision, preserving the existing results. Report claims must remain restricted to the mathematical and empirical evidence actually obtained.

References

[1] Advay. Historical pelt data, provenance and observation convention. Team repository, revision `28dbe1e`, files `data/historical/README.md`, `provenance.json`, `split.json` and `docs/advay_observation_convention.md`, 2026. Reviewed 28 October 2026. `https://github.com/FriedelCraft/lynx-hare-project/tree/28dbe1e2a611719f3d65e5922db528ae423724f2`.

[2] Bob Carpenter. Predator–prey population dynamics: The Lotka–Volterra model in Stan. Stan case study, 28 January 2018. `https://mc-stan.org/learn-stan/case-studies/lotka-volterra-predator-prey.html`, 2018. Accessed 26 October 2026. Public transcription and source discussion for the 1900–1920 pelt series; quantities in thousands of pelts.

[3] C. S. Holling. Some characteristics of simple types of predation and parasitism. *The Canadian Entomologist*, 91(7):385–398, 1959. `https://doi.org/10.4039/Ent91385-7`.

[4] Joel. Equilibria, stability and hopf bifurcation. Team report contribution, revision `7d17deee`, 2026. Accessed 26 October 2026. `https://github.com/FriedelCraft/lynx-hare-project/blob/7d17deee1cb685f016ff16cb6706e8db12ebe6b2/report/joel_stability_section.tex`.

[5] Lynx Hare Project team. Lynx hare project: Shared model, synthetic configuration and Yash's PINN method. GitHub repository, revision `7d17deee`, 2026. Accessed 26 October 2026. `https://github.com/FriedelCraft/lynx-hare-project/tree/7d17deee1cb685f016ff16cb6706e8db12ebe6b2`.

[6] Rishi. Numerical simulations: Team handoff. Team repository, revision `28dbe1e`, file `docs/rishi_simulation_handoff.md`, 2026. Reviewed 28 October 2026. `https://github.com/FriedelCraft/lynx-hare-project/tree/28dbe1e2a611719f3d65e5922db528ae423724f2`.

[7] Michael L. Rosenzweig and Robert H. MacArthur. Graphical representation and stability conditions of predator–prey interactions. *The American Naturalist*, 97(895):209–223, 1963. `https://doi.org/10.1086/282272`.

[8] Yash and Lynx Hare Project team. Historical fitting configuration, implementation and handoff. Team repository, revision `28dbe1e`, files `configs/historical.json`, `src/lynx_hare/historical_pinn.py`, `model.py`, `pinn.py` and `docs/yash_historical_handoff.md`, 2026. Reviewed 28 October 2026. `https://github.com/FriedelCraft/lynx-hare-project/tree/28dbe1e2a611719f3d65e5922db528ae423724f2`.